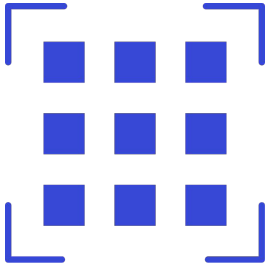
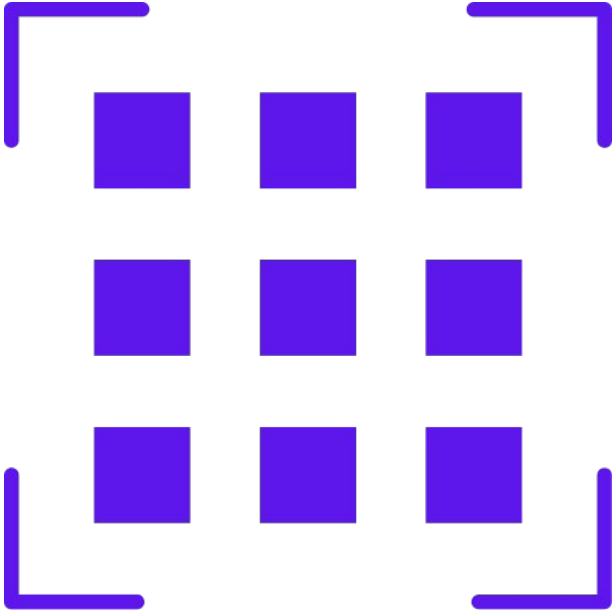
Array/Lists





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Pre-requisites

- Basic Python Knowledge
- Time and Space Complexity
- Willingness to learn

What are Data Structures?

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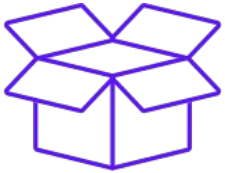
Data Structures

- Particular way of organizing, and storing the data
- Tools to build efficient algorithms
- The right one depends on the task
- Common Data Structures:
  - Arrays
  - Linked Lists
  - Stacks
  - Queues
  - Trees
  - Maps(dictionaries)
  - Sets
  - Tries, etc...



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Arrays

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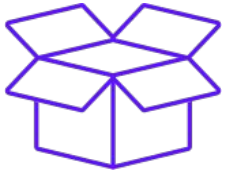
Arrays

- Collection of items stored at contiguous memory locations.
- Storing multiple items of the same type together.
- Easier to calculate the position of each element by simply adding an offset to a base value.

Array members →	x[0]	x[1]	x[2]	x[3]	x[4]	x[5]
	1	10	8	1	9	1
Array indices →	0	1	2	3	4	5



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Static Arrays

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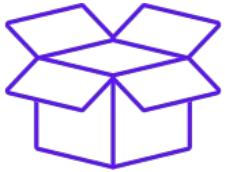
Static Arrays

- A type of array in which the size or length is determined when the array is created and/or allocated.

	7	3	6	
	*0	*4	*8	



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Static Arrays

Value		1	3	5
Address		*0	*4	*8

Value		a	b	c
Address		*0	*1	*2

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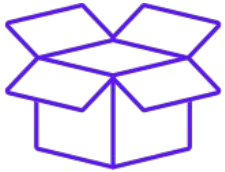
Static Arrays - Reading

- Reading is Very fast : O(1)
- Why?

```
print(my_array[4])
```



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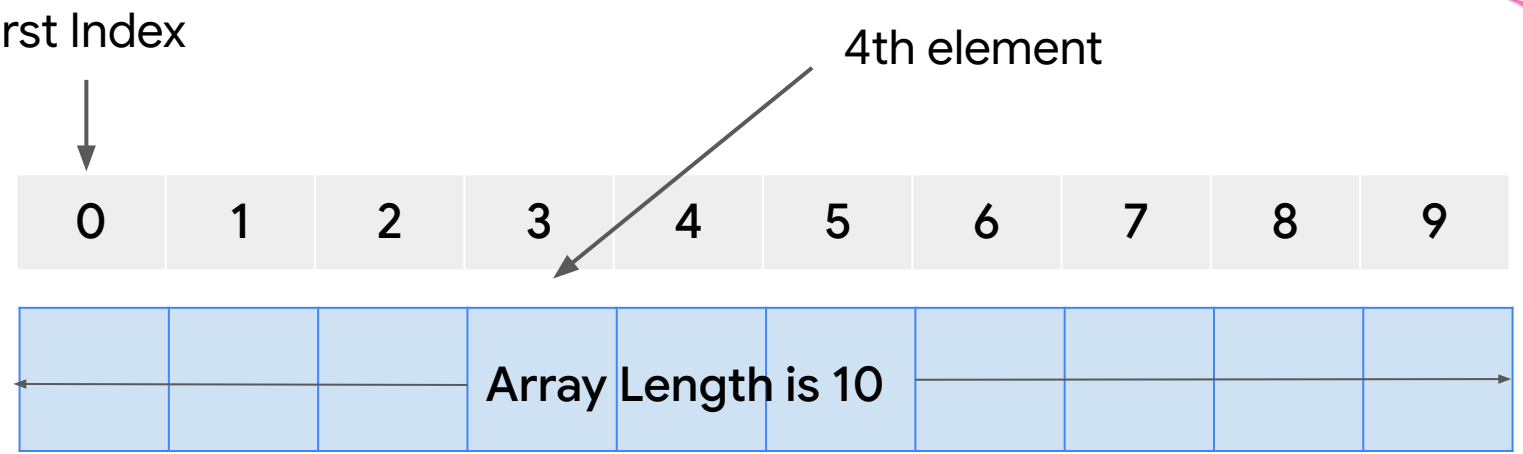
Static Arrays - Reading



Each integer occupies 4 bytes, so let us say the 0th index is at memory location 1400

1400 + (3 * 4) = 1412

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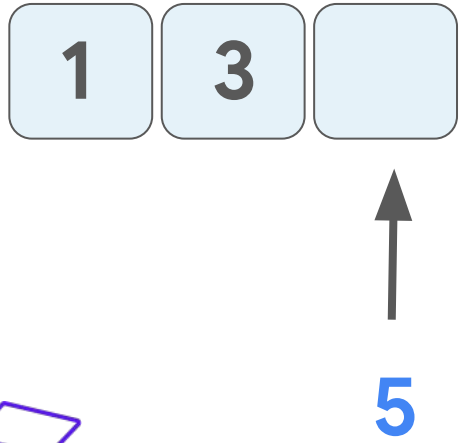
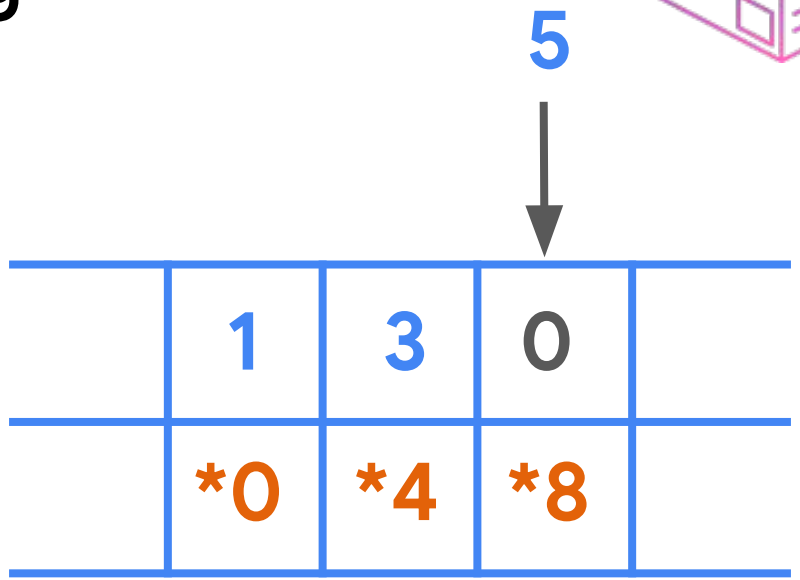
Static Arrays - Writing

- Writing is an instant operation





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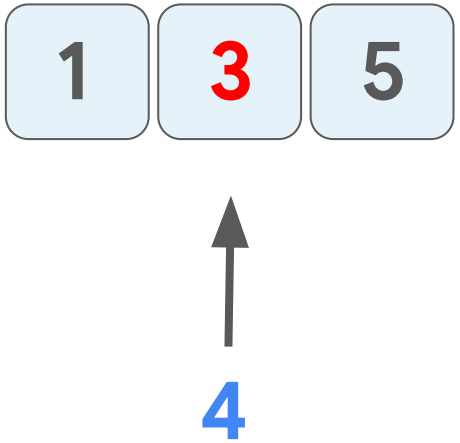
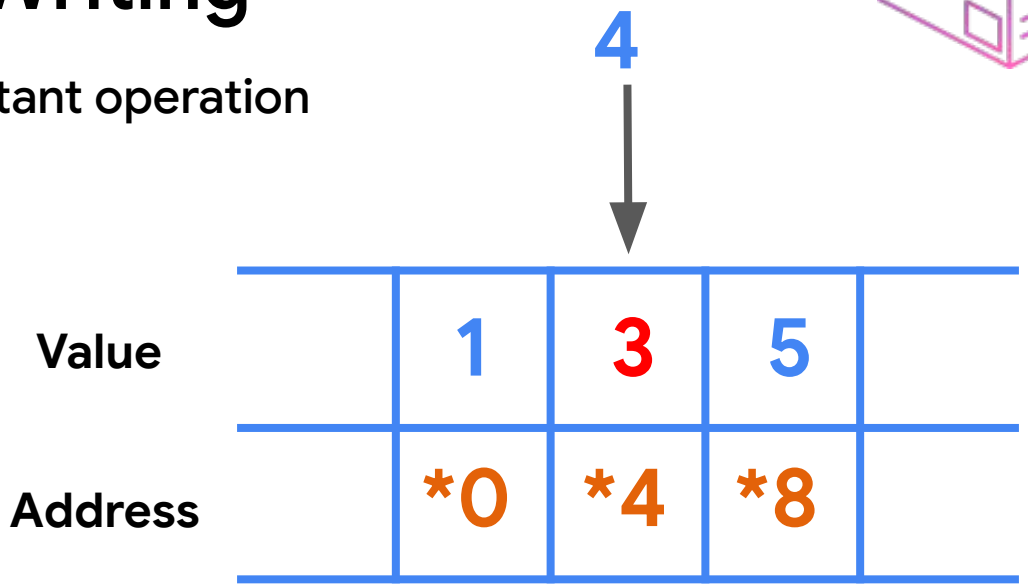
Static Arrays - Writing

- Overwriting is an instant operation





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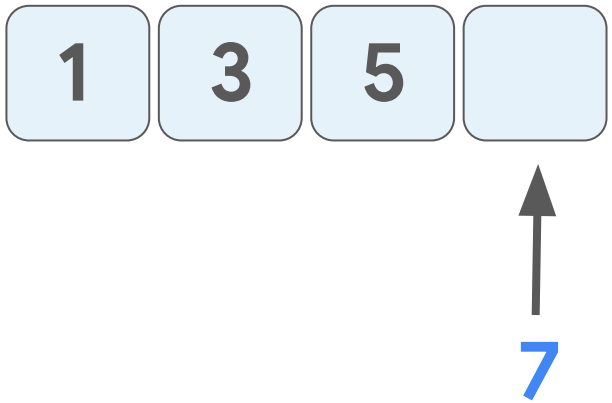
Static Arrays - Writing

- What happens when we want to add an element to our array but don't have any empty spots?



Value		1	3	5	
Address		*0	*4	*8	

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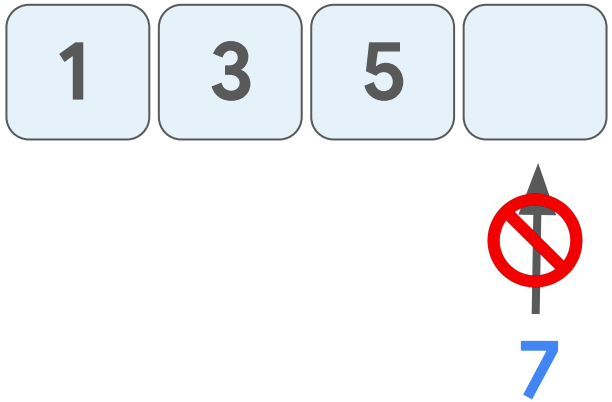
Static Arrays - Writing

- The operating system might be using this part of the memory for some other purpose.
- So we are not necessarily allowed to put our seven there.



Value	1	3	5	⊘
Address	*0	*4	*8	

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Static Arrays - Summary

Operation	Big-O Time
Read i-th Element	O(1)
Write i-th Element	O(1)

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Dynamic Arrays

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Dynamic Arrays

- A Dynamic array has the ability to resize itself automatically when an element is inserted or deleted.

- Double the size when it reaches capacity, cut by half when it’s down to quarter

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Does Python list allow the storage of different types ?

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Dynamic Arrays

- No way to find out arr[3] without traversing the full list

```
arr = [3, 0.5, 'a', "1"]
```

Value		3	0.5	a	"1"	
Address		*0	*4	*12	*13	

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Dynamic Arrays

- A list in Python is implemented as an array of references.

- what you are actually creating is an array of references like :

0 1 2 3

[0xa3d25342, 0x635423fa, 0xff243546, 0x2545fade]

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Dynamic Arrays

- Each element is a reference that "points" to the respective objects in memory

		*0	*1	*2	*3	
Reference		*10	*14	*22	*23	
Value		3	0.5	a	“1”	

- **Memory Address**: A unique identifier for a specific location in memory where data is stored, typically represented as a hexadecimal value.
- **Reference**: A variable that acts as an alias for another object, allowing indirect access to that object without directly storing its memory address.

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Dynamic Arrays

- This works because the all references are of the same size unlike the actual values they point to.

	*0	*1	*2	*3	
Reference	*10	*14	*22	*23	
Value	3	0.5	a	“1”	

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Common Terminologies

- Pushing/Appending : inserting an element at the next empty spot

- Popping: removing the element from the end of the list

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Pushing

- Whenever we reach our array's capacity, our interpreter allocates another memory which is double our original capacity and copies over all the old elements with now more space.

- Isn't this operation costly?

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Pushing

- It takes amortized O(1) time! Let's see why

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Pushing

- Our goal is to push 8 elements to a dynamic array.
- Assumption: Our initial capacity is of size one.

5	6	7	8	9	1	2	3

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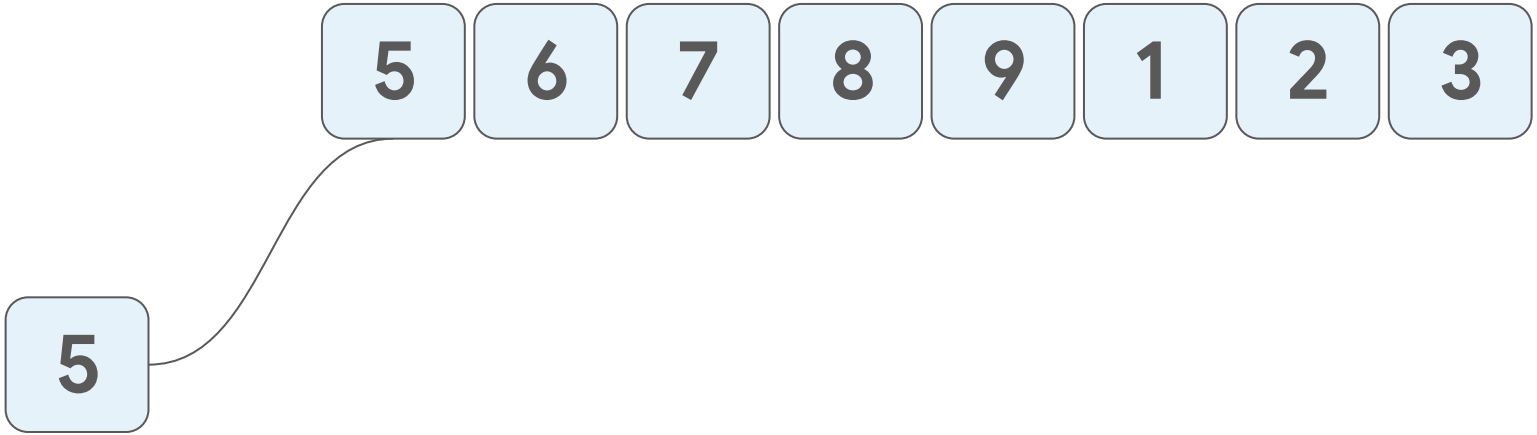
Pushing

- After pushing 5, no place for 6



Operation count: 1

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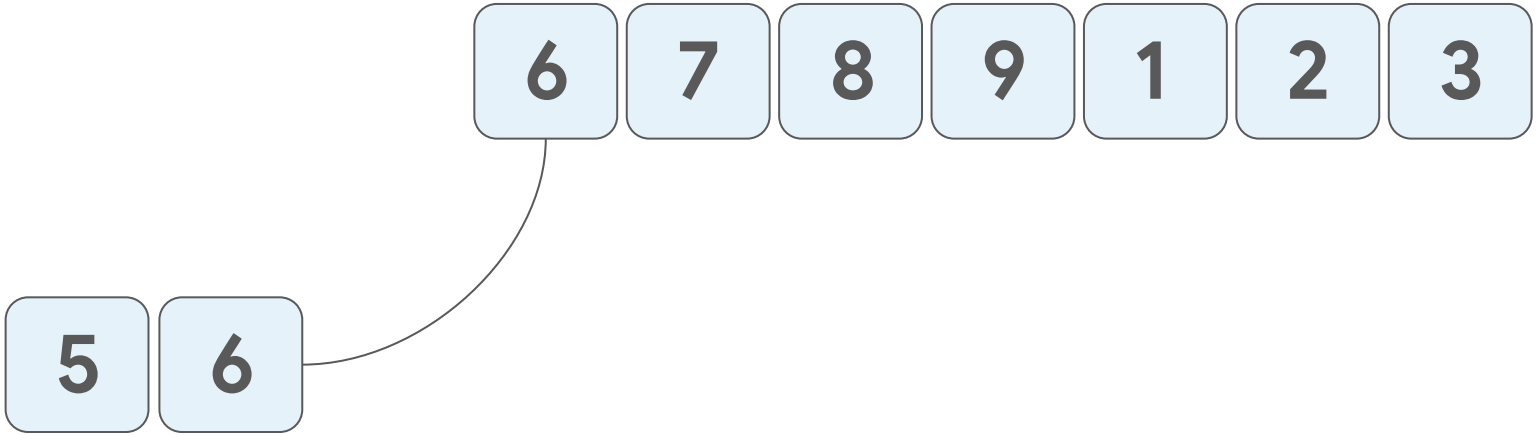
Pushing

- Copy the original array ([5]) to the new allocated memory which is double the old size and push 6 onto it



Operation count: 1 + 2

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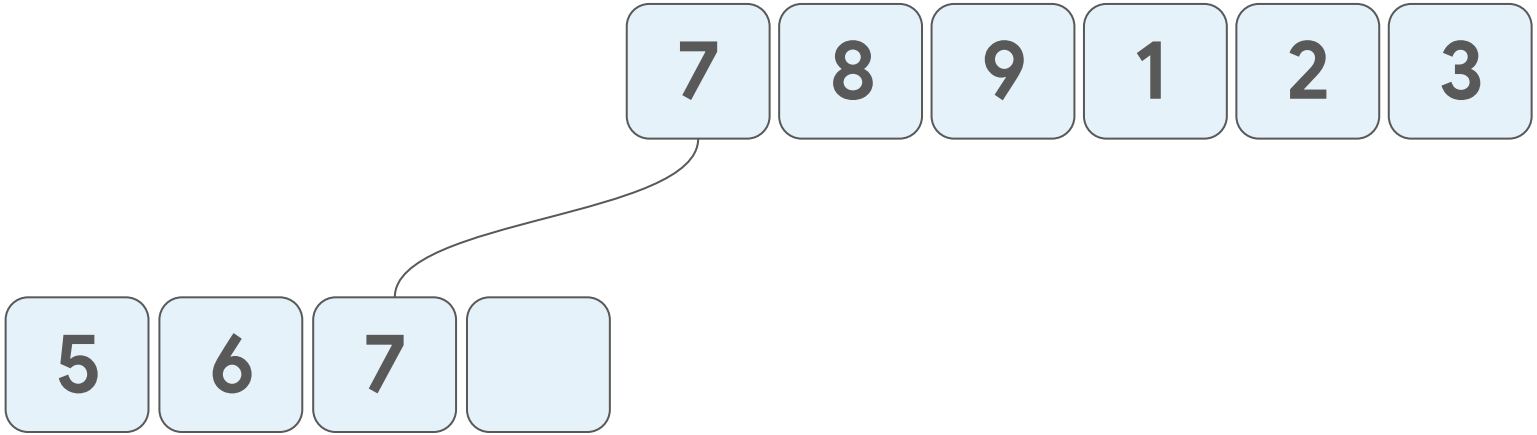
Pushing

- Allocate a new array of size 4 and copy over all the old elements and then we append 7



Operation count: 1 + 2 + 3

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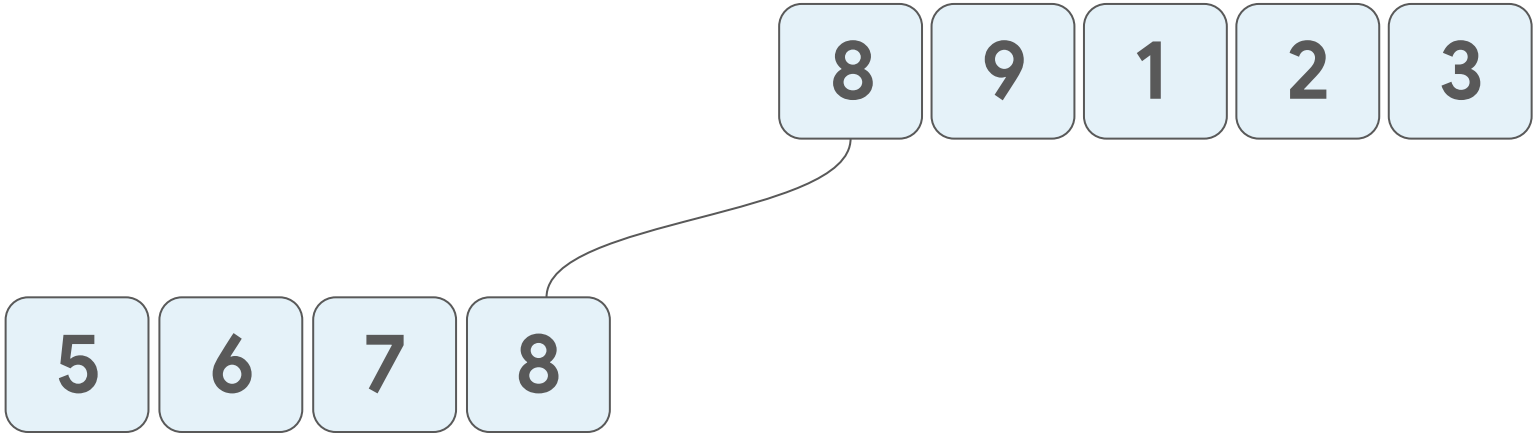
Pushing

- Push 8 onto the array. Now we have run out of space for the next elements.



Operation count: 1 + 2 + 4

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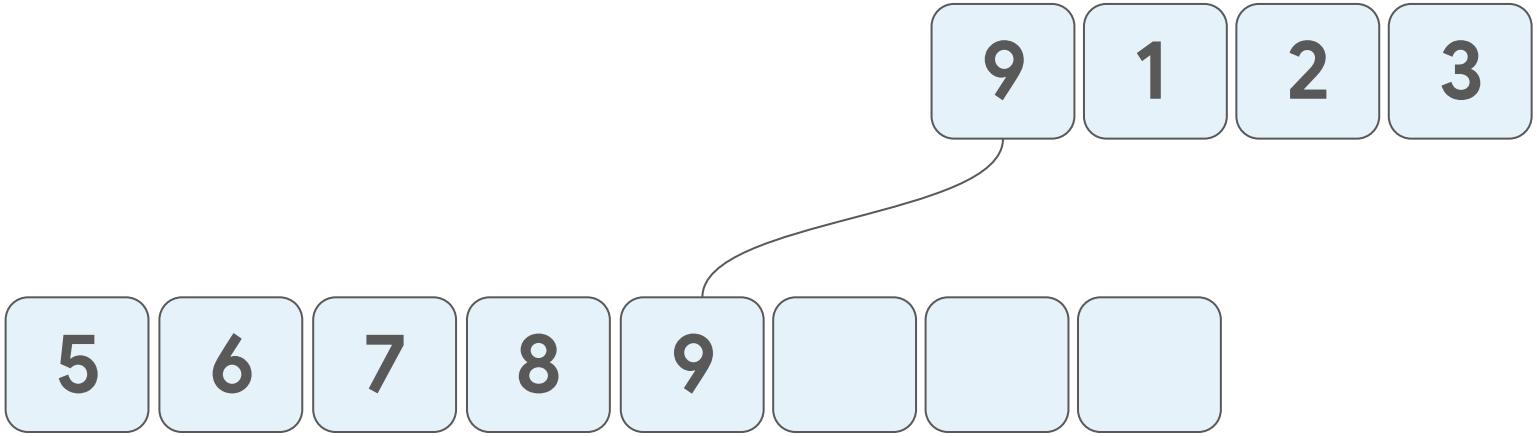
Pushing

- Allocate a new array of size 8 and copy over all the old elements and then we append the element 9



Operation count: 1 + 2 + 4 + 5

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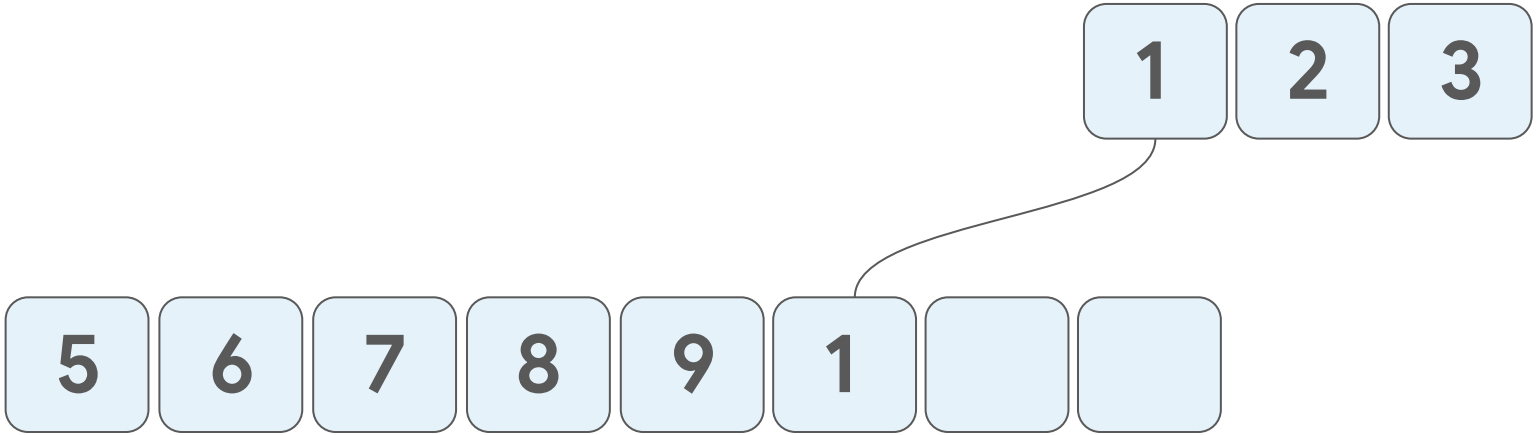
Pushing

- Push/append 1 into the array



Operation count: 1 + 2 + 4 + 6

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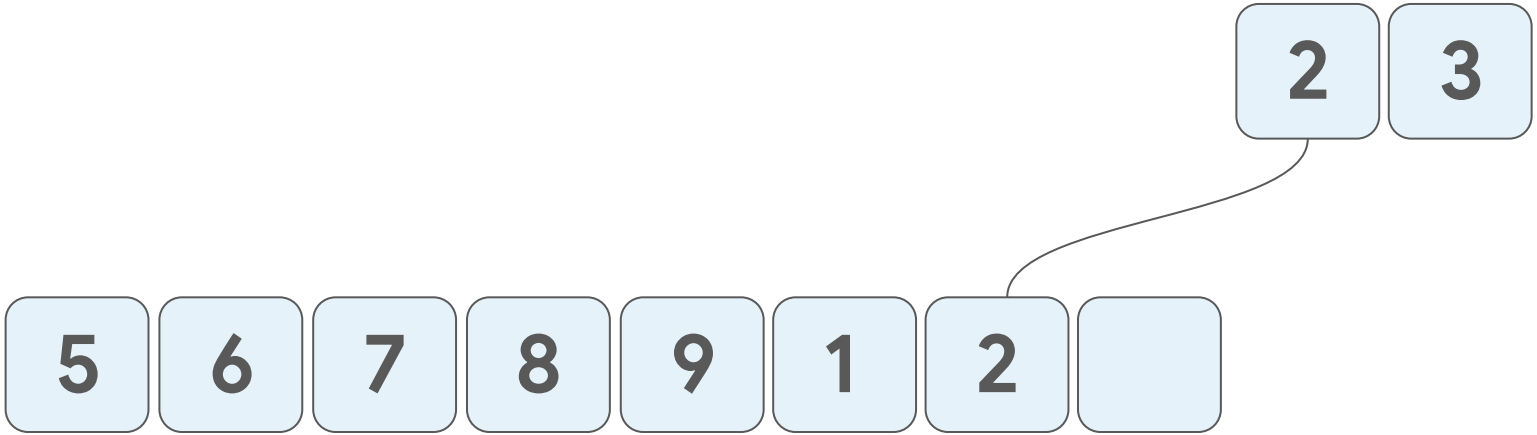
Pushing

- Push/append 2 into the array



Operation count: 1 + 2 + 4 + 7

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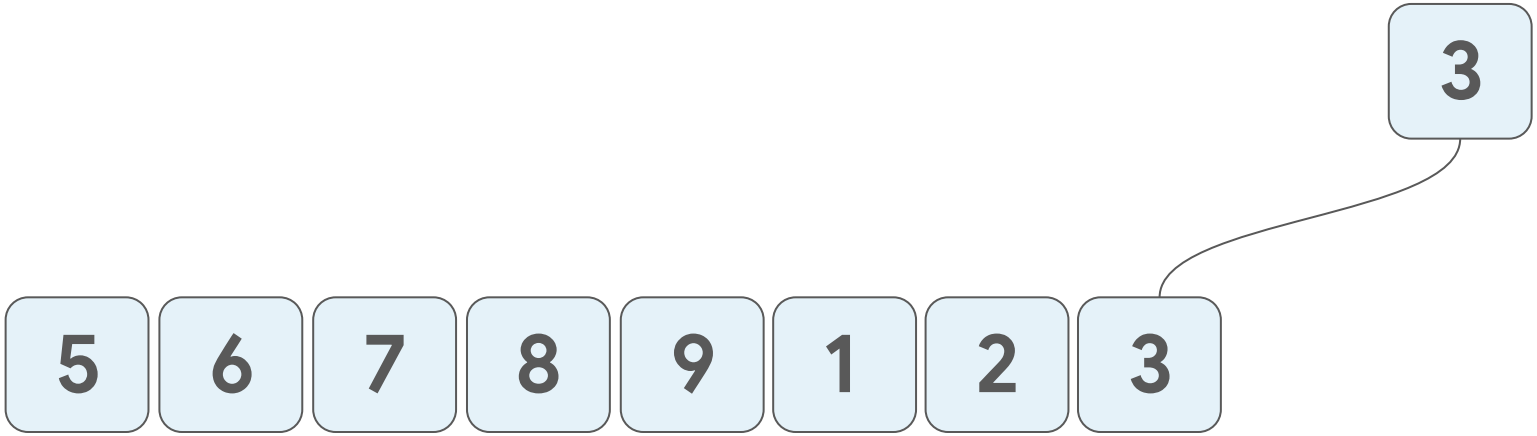
Pushing

- Push/append 3 into the array



Operation count: 1 + 2 + 4 + 8

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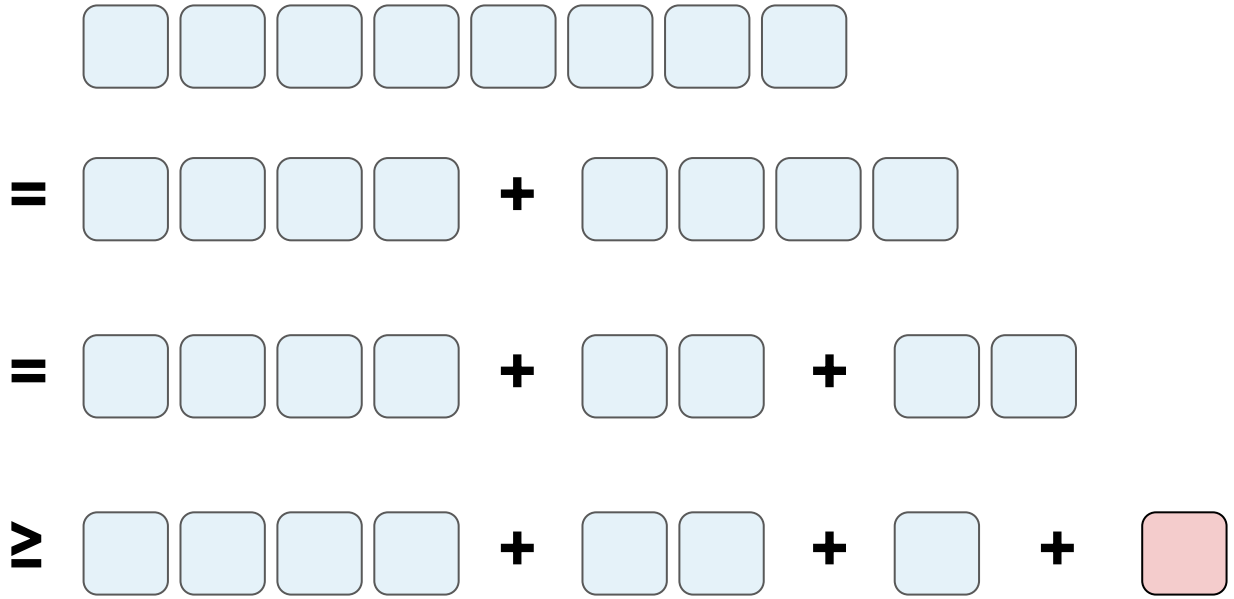
Pushing- Power Series

- For a length of 8, it took us 1 + 2 + 4 + 8 operations.
- The last summand always dominates the sum.



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Pushing - Power Series

- For a length of 8, it took us 1 + 2 + 4 + 8 operations.
- The last summand always dominates the sum.

$$2 \geq 1$$
$$4 \geq 1 + 2$$
$$8 \geq 1 + 2 + 4$$
$$\cdot$$
$$\cdot$$
$$\cdot$$
$$2*N \geq 1 + 2 + 4 + 8 + \ldots + N$$

- Thus to push N elements, it will take us no more than 2N operations, making the time complexity O(N).

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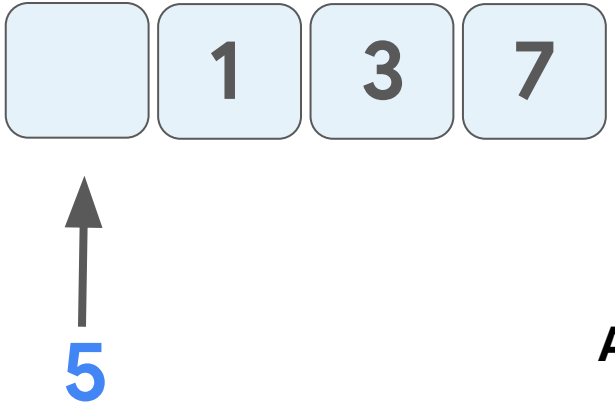
Pushing - Power Series

- This means, to push a single element onto an array, on average, we say it is an instant - O(1) - operation.
- Because after a while, the copying operation becomes very infrequent.
- Hence, pushing an element onto an array is Amortized O(1) operation.

Dynamic Arrays - Insertion

- Inserting an element in an arbitrary position is, in the worst case, an O(n) operation.



Value		1	3	7
Address	*0	*4	*8	*12

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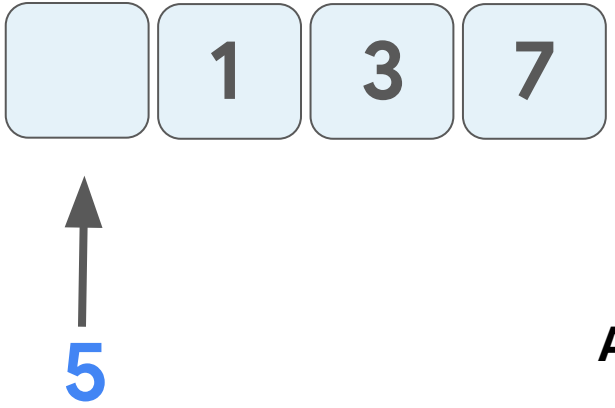
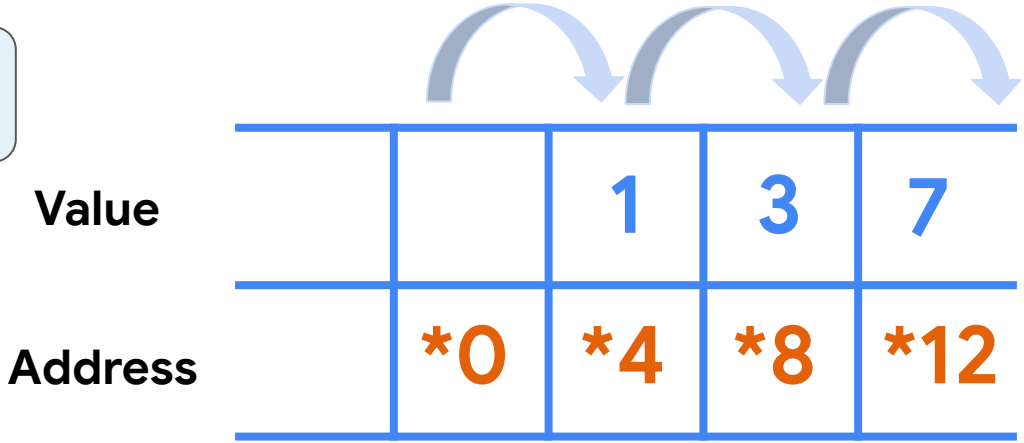
Dynamic Arrays - Insertion

- We need to shift all elements to the right of the new element, to not lose data, and create the empty space needed.





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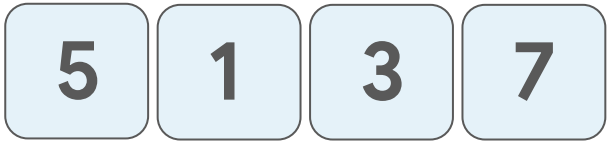
Dynamic Arrays - Insertion

- The shifting is what costs us time.



Value	5	1	3	7
Address	*0	*4	*8	*12

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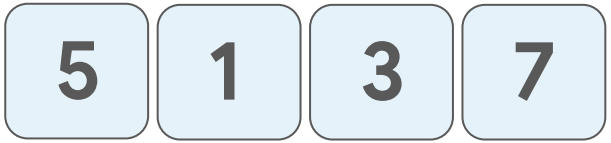
Dynamic Arrays - Removal

- What about removing an element?



Value	5	1	3	7
Address	*0	*4	*8	*12

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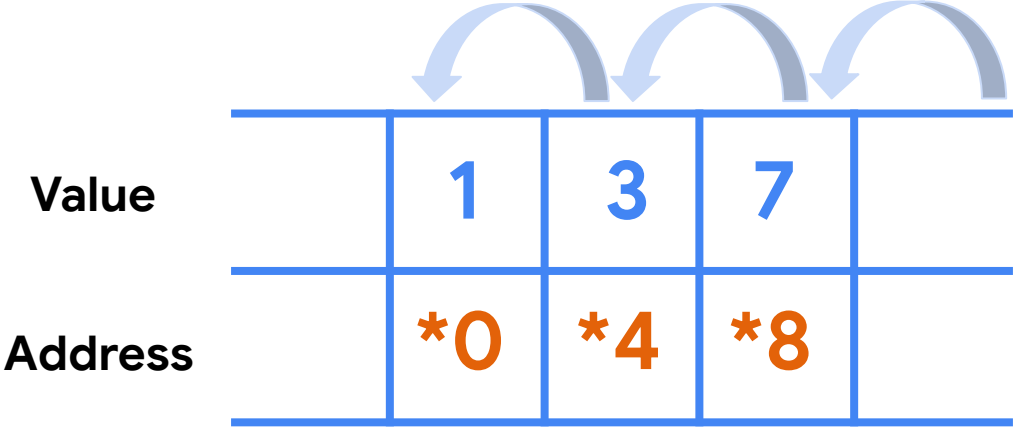
Dynamic Arrays - Removal

- It is O(n) at worst.



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Checkpoint

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Traversing Lists

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Traversing Lists - Reverse Traversal

Approach

- Start from the last index
- Decrement by 1 until -1

Implementation

```
for index in range(len(array)-1, -1, -1):
    print(array[index])
```

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Traversing Lists - Even/Odd Indices Traversal

Approach

- Start from the first even/odd index
- Increment by 2

Implementation

```
step = 2 # incremental value
# even indices traversal
size = len(array)
for index in range(0, size, step):
    print(array[index])
```

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Traversing Lists - Circular Traversal

Approach

- Start from first index to size*2
- Incrementally go 1 step but modulo index by size

Implementation

```
size = len(array)
for index in range(0, 2 * size):
    value = array[index % size]
    print(value)
```

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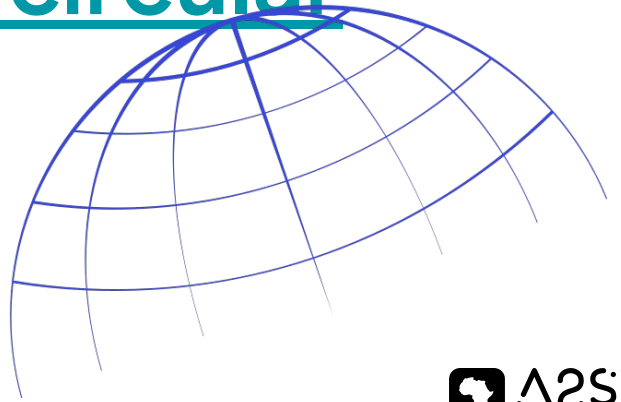
Pair Programming

Find the winner of the circular game



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Swapping Elements

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Swapping Elements

Approach

- Approach 1
  - Store first element in a variable
  - Change first element with second element
  - Change second element with stored value
- Approach 2 : Use tuple unpacking to swap

Implementation

```
# approach 1
temp = arr[i]
arr[i] = arr[j]
arr[j] = temp

# approach 2
arr[i], arr[j] = arr[j], arr[i]
```

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Storing letter and Numbers

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Storing - Using Arrays as Dictionaries

- For Letters
  - Character's ASCII value can be the index by correcting the offset by 65 ('A') for uppercase letters and 97 ('a') for lower case letters

- For Numbers
  - The number itself is the key

- Note: the advantage of this approach is that you can use built in functions like `sum`, `sort`, `max`, `min` on the frequency list whereas maps don't give you this flexibilities.

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Storing - Implementation

```
word = "abceabce"
arrayDictionary = [0]*26
offset = ord('a') # 'a' has ASCII value of 97
for char in word:
    ascii = ord(char)
    arrayDictionary[ascii - offset] += 1

print(arrayDictionary)
# [2, 2, 2, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
#  a  b  c  d  e  f  g  h  i  j  k  l  m  n, o  p  q  r  s  t  u  v  w  x  y  z
```

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Pair Programming

Reverse string

Apply operations to an array

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Resources

- https://neetcode.io/courses/dsa-for-beginners/0: useful on laying out the fundamentals
- https://www.geeksforgeeks.org/python-arrays/: covers large range of topics for python
- Python List pop() Method - GeeksforGeeks: good examples
- Python List insert() Method With Examples - GeeksforGeeks: good examples
- Python | Which is faster to initialize lists? - GeeksforGeeks: good article on list initialization

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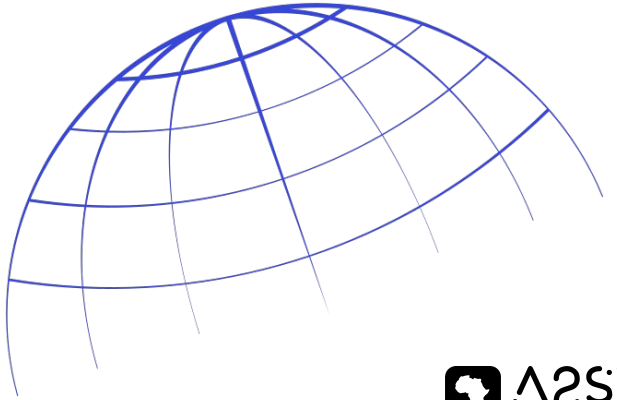
Quote of the Day

If you want to enjoy the rainbow, be prepared to endure the storm

Warren W. Wiersbe



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